

Simon Malnar

smalnardev@gmail.com

Software engineer for over 12 years with experience in partnering with scientists, researchers and industry experts to develop software solutions that communicate and operationalise complex information. Delivered high quality projects spanning data visualisation, interactive learning, commercial gaming and low-latency networking. Skilled in building custom tooling to optimise operational efficiency and AI-driven workflows with a strong foundation in full-stack web development through TypeScript and PHP.

TECHNICAL SKILLS & COMPETENCIES

- **Languages & Frameworks:** C#, JavaScript, TypeScript, Node.js, Python, C++, Java, PHP, SQL, Postgres, React
- **Engines & Tools:** Unity, Git, npm, MRTK, OpenXR
- **Cloud & Infrastructure:** GCP, Firebase, AWS, CI/CD Pipelines, REST APIs, Docker

PROFESSIONAL EXPERIENCE

Oh Baby Games, (Remote) — *Lead Developer*

December 2022 - June 2026

- Led the development of 2 commercial titles through their full lifecycle, taking them from prototype to release.
- Contributed core system programming and technical support across several concurrent projects to support cross-functional teams and accelerate development.
- Built custom Unity Editor tooling and AI workflows to simplify the design and implementation process and reduce feature iteration times for designers.

CSIRO, Black Mountain, ACT — *Research Engineer*

November 2016 - December 2022

- Rapidly mastered diverse business domains to pioneer custom XR software solutions across various fields including agriculture, entomology, urban planning and genomics.
- Researched and architected low-latency networking systems for inter-device connectivity and multi-user synchronisation across XR environments.
- Translated specialised scientific requirements and massive datasets into intuitive, interactive 3D data visualisation applications.
- Developed and optimised software applications tailored to specific hardware platforms including the Microsoft HoloLens, HTC Vive and Oculus Rift.

Offpeak Games, Perth (Remote) — *Programmer*

June 2016 - June 2017

- Co-developed and shipped 3 commercial multiplayer VR titles on the Steam platform: Crazy Fishing, Arcane and Valiant
- Designed and implemented unique 3D physics-based interaction systems specifically optimised for the tracking capabilities and hardware constraints of the HTC Vive and Oculus Rift
- Programmed and synchronised networked AI behaviours, resolving latency and state consistency challenges to ensure responsive enemy logic in multiplayer sessions

Gitsumm Inc., (Remote) — *Programmer*

January 2014 - January 2015

- Engineered and deployed 6 cross-platform HTML5 web games utilising TypeScript and the Phaser framework
- Collaborated in an agile team environment to build and optimise multi-platform games using C# and the Unity engine
- Developed secure server-side infrastructure using PHP, architecting global leaderboard systems, custom administrative dashboards and user moderation/blacklist tools.

EDUCATION

University of Canberra, ACT, Australia — *Bachelor of Software Engineering*

2013 - 2016

- Completed with a GPA of 6.8.
- Earned "Microsoft prize for Outstanding Graduating Student in the course of Bachelor of Software Engineering", "ACS Bachelor of Software Engineering First Year Prize" and "Elsie Hope Solly Prize"

Academy of Interactive Entertainment, ACT, Australia — *Completed Certificate II and III in Information, Digital Media and Technology*

2011 - 2013

- Earned "AIE prize for Mathematics and Information Technology"